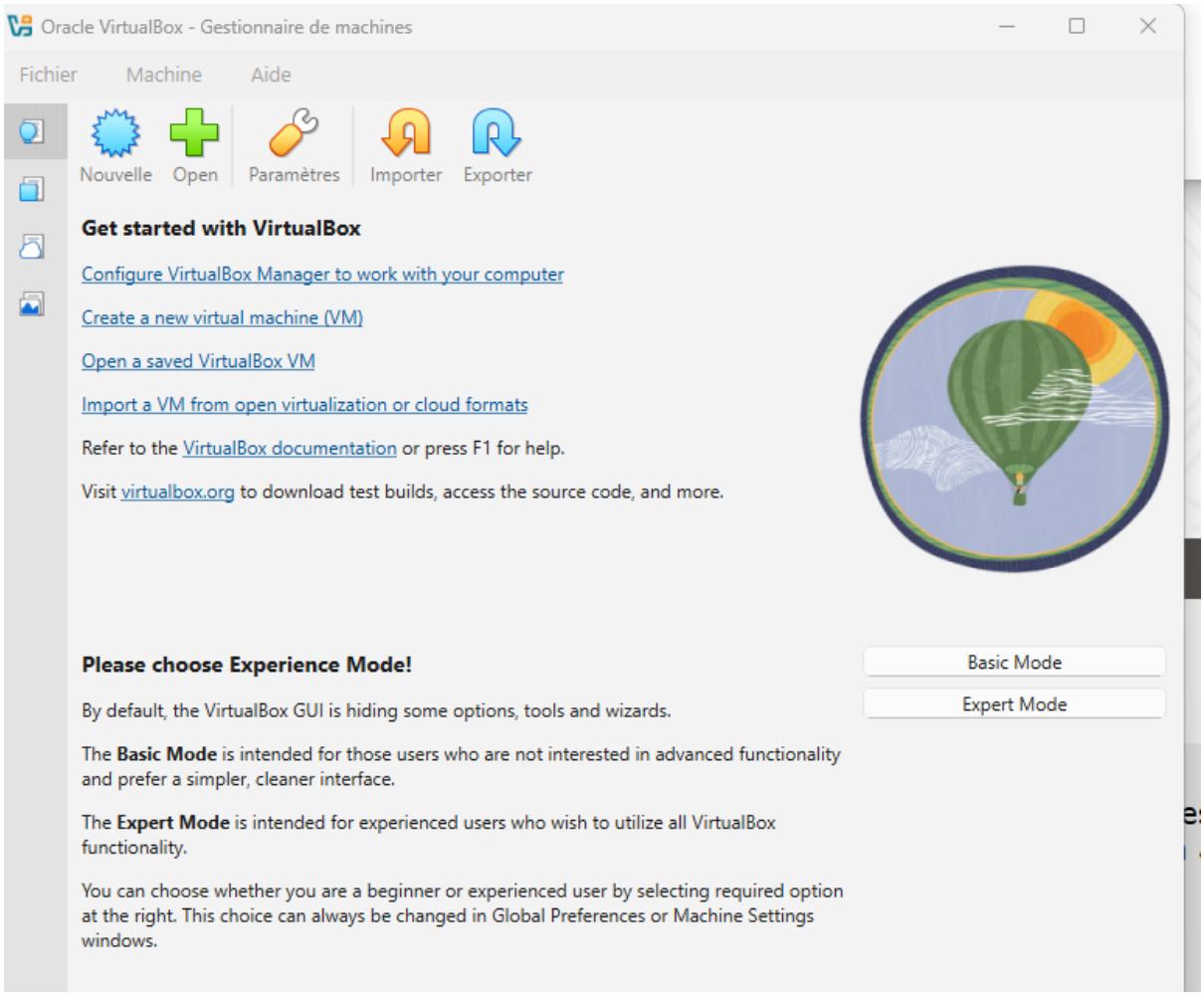
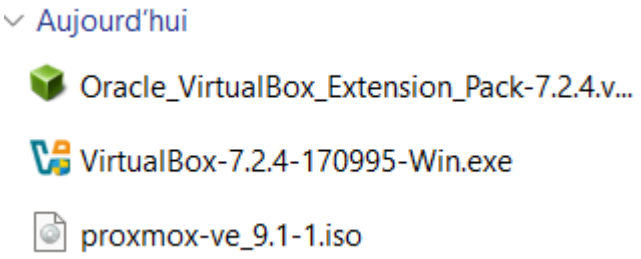
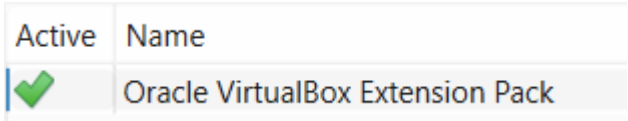
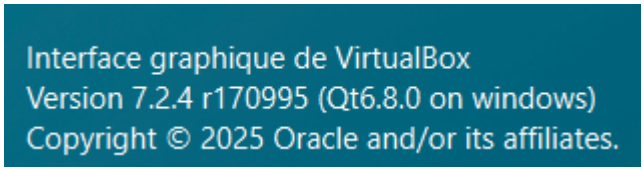
Documentation VM Semaine 1 :

Installation de VirtualBox, de son extension pour la version correspondante (7.2.4), et de l'ISO Proxmox pour l'installation des machines virtuelles et de son environnement.







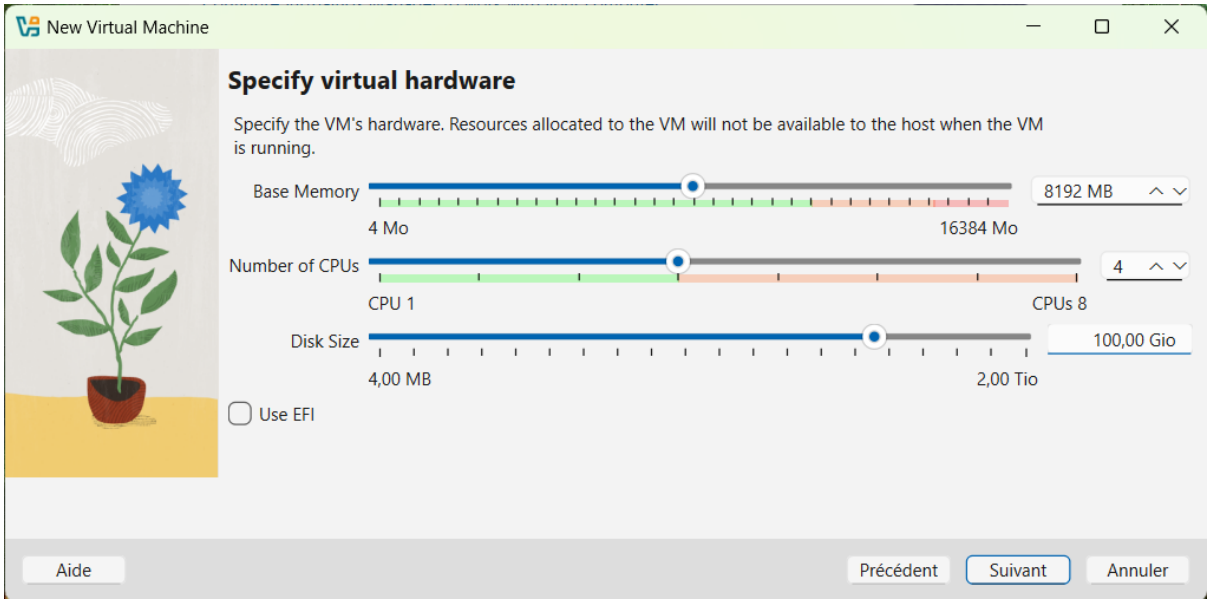


Installation Proxmox sur VirtualBox :

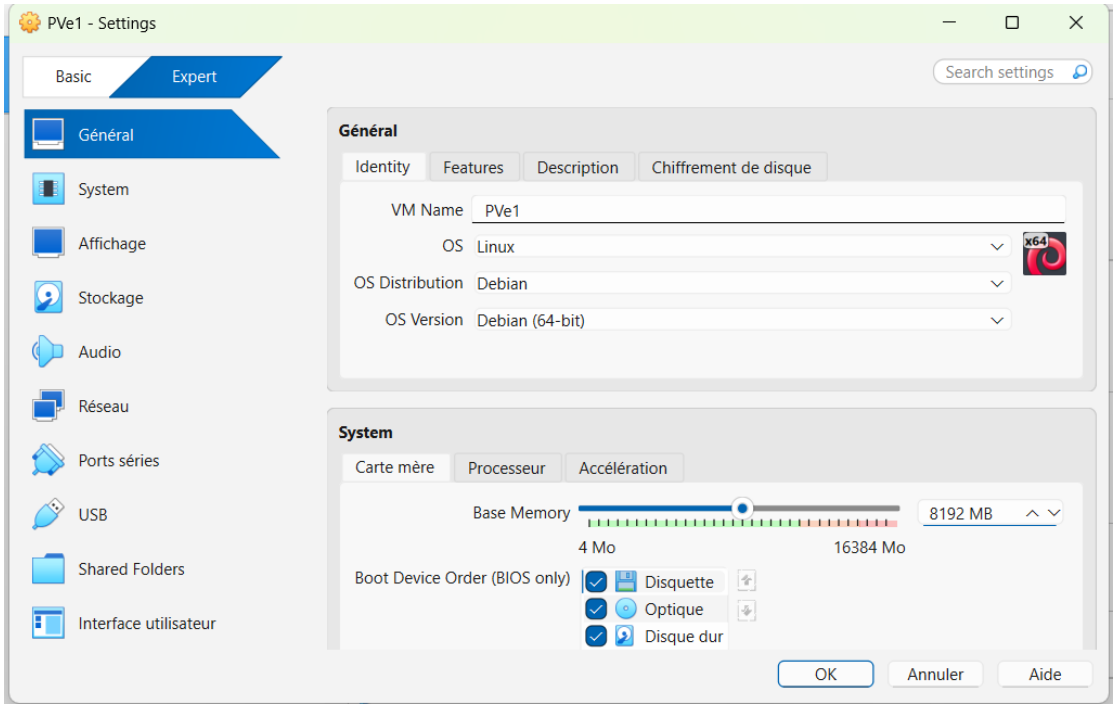
Proxmox est une plateforme qui sert à gérer les machines virtuelles ou des conteneurs, qui est basé sur Linux, plus précisément Debian.

Création PVE :

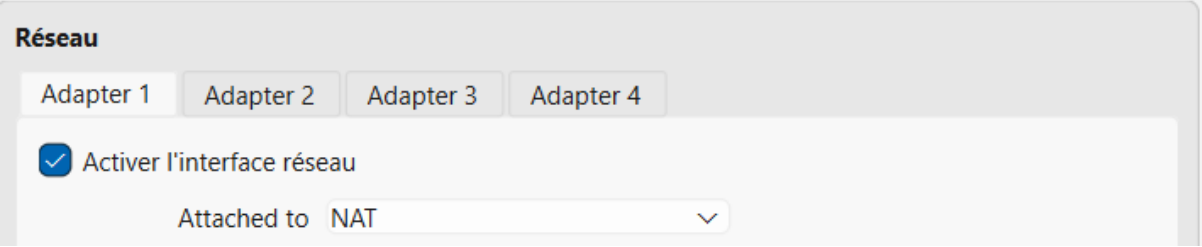
Je viens de faire une configuration du système (taille du disque dur, nombre de processeur)

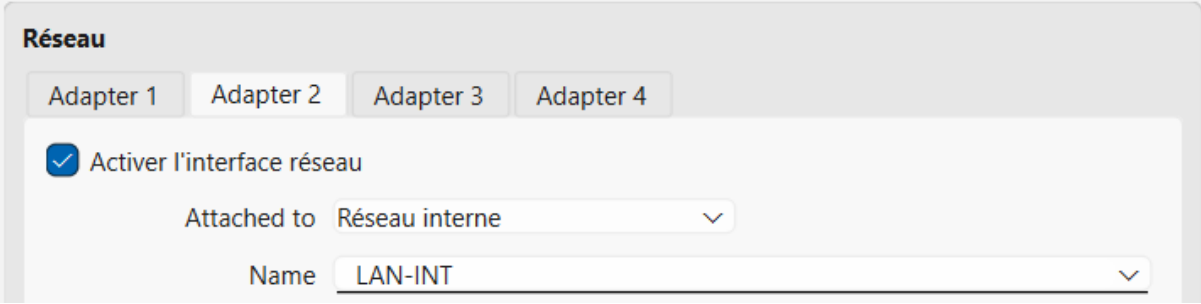


Ici voici les informations générales de la VM (les mêmes pour la PVE2):



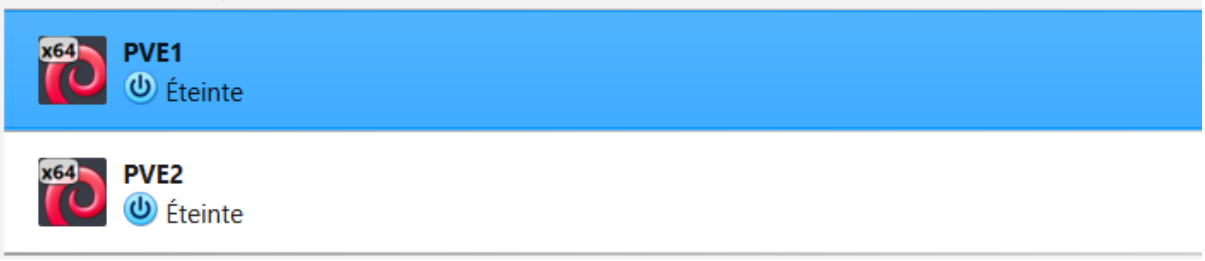
Puis la configuration du réseau (2 interfaces réseau)





Faire la même pour la PVE2

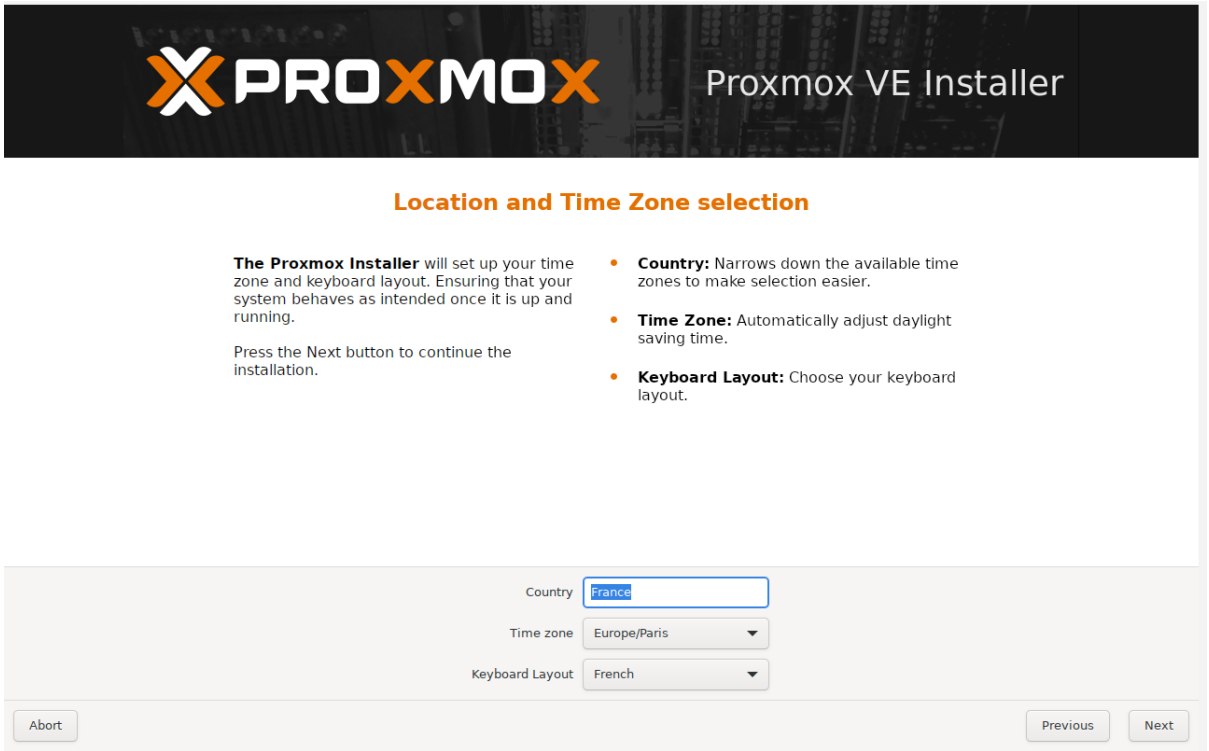
Les deux VM sont bien configurées.



<u>Installation PVE :</u>

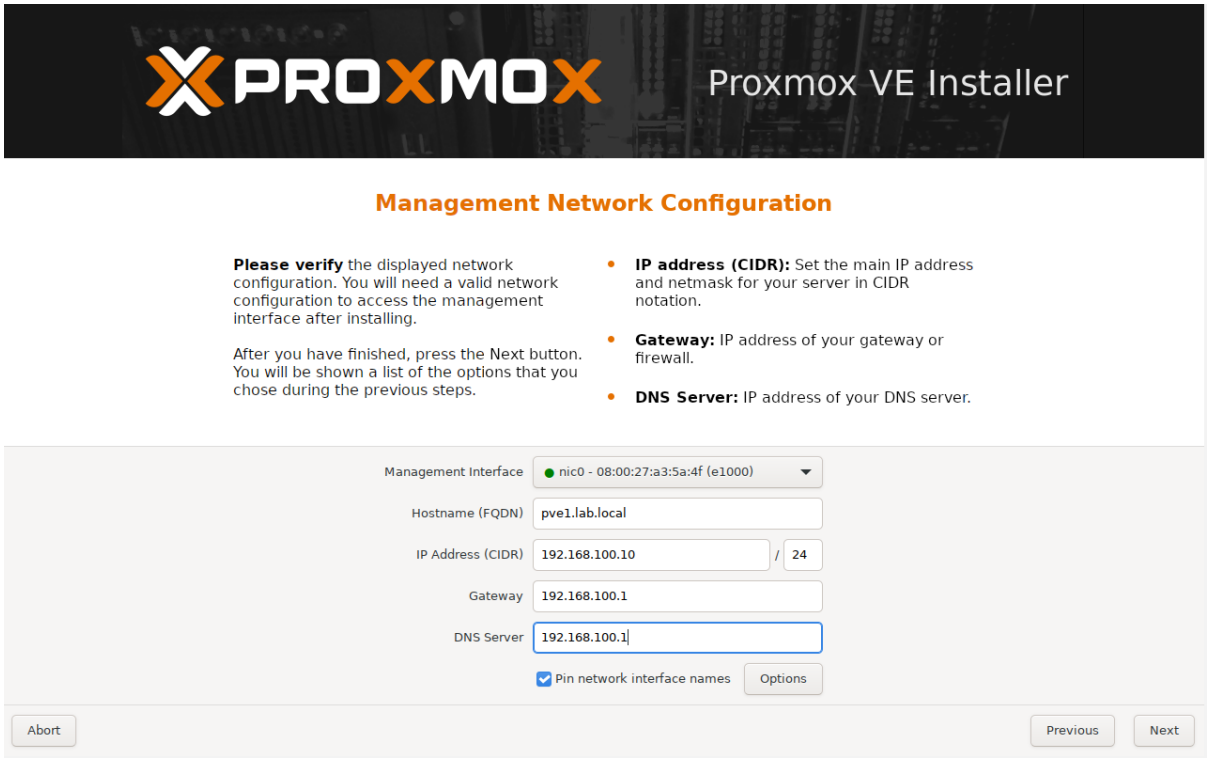
On choisit l’emplacement de l’environnement et son disque dur.

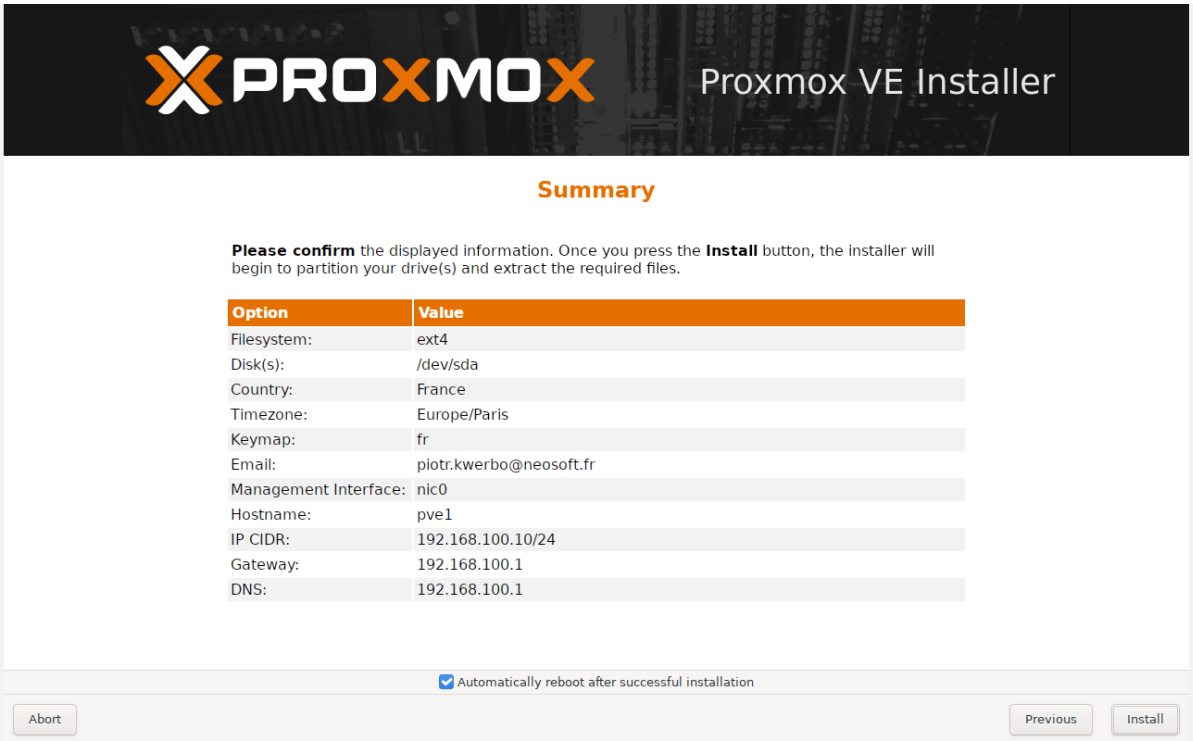
On choisit notre localisation, le fuseau horaire et le clavier qu'on utilise (dans ce cas c'est France)



On configure notre adresse IP, la passerelle et le DNS (dans ce cas la passerelle et le DNS c'est le même).

On utilise cette adresse pour avoir accès à l'interface Web de Proxmox





Option	Value
Filesystem:	ext4
Disk(s):	/dev/sda
Country:	France
Timezone:	Europe/Paris
Keymap:	fr
Email:	piotr.kwerbo@neosoft.fr
Management Interface:	nic0
Hostname:	pve1
IP CIDR:	192.168.100.10/24
Gateway:	192.168.100.1
DNS:	192.168.100.1

Résultat après l’activation de la VM :

```
--------------------------------------------------------------------------------

Welcome to the Proxmox Virtual Environment. Please use your web browser to
configure this server - connect to:

  https://192.168.100.10:8006/

--------------------------------------------------------------------------------

pve1 login: _
```

La même pour la PVE2

```
--------------------------------------------------------------------------------

Welcome to the Proxmox Virtual Environment. Please use your web browser to
configure this server - connect to:

  https://192.168.100.11:8006/

--------------------------------------------------------------------------------

pve2 login: _
```

Invite de commande Proxmox

```
pve1 login: root
Password:
Linux pve1 6.17.2-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.2-1 (2025-10-21T11:55Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@pve1:~# pvecm create cluster-bts
```

<u>Login</u> : root

<u>Mdp</u> : Admin+-12

```
pve2 login: root
Password:
Linux pve2 6.17.2-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.2-1 (2025-10-21T11:55Z) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@pve2:~#
```

<u>Login</u> : root

<u>Mdp</u> : Admin+-11

Pour voir l'IP de la machine, c'est la commande Linux classique : *ip a*

```
root@pve1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr1 state UP group default qlen 1000
    link/ether 08:00:27:a3:5a:4f brd ff:ff:ff:ff:ff:ff
    altname enx080027a35a4f
3: nic1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 08:00:27:ac:2e:37 brd ff:ff:ff:ff:ff:ff
    altname enx080027ac2e37
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:ac:2e:37 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feac:2e37/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
5: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:a3:5a:4f brd ff:ff:ff:ff:ff:ff
    inet6 fd17:625c:f037:2:a00:27ff:fea3:5a4f/64 scope global dynamic mngtmpaddr proto kernel_ra
       valid_lft 86169sec preferred_lft 14169sec
    inet6 fe80::a00:27ff:fea3:5a4f/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

```
root@pve2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr1 state UP group default qlen 1000
    link/ether 08:00:27:05:6b:14 brd ff:ff:ff:ff:ff:ff
    altname enx080027056b14
3: nic1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 08:00:27:ba:c4:d3 brd ff:ff:ff:ff:ff:ff
    altname enx080027bac4d3
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:ba:c4:d3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.11/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feba:c4d3/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
5: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:05:6b:14 brd ff:ff:ff:ff:ff:ff
    inet6 fd17:625c:f037:2:a00:27ff:fe05:6b14/64 scope global dynamic mngtmpaddr proto kernel_ra
       valid_lft 86253sec preferred_lft 14253sec
    inet6 fe80::a00:27ff:fe05:6b14/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Pour voir les routes des IP : *ip route*

```
root@pve1:~# ip r
192.168.100.0/24 dev vmbr0 proto kernel scope link src 192.168.100.10
```

```
root@pve2:~# ip route
192.168.100.0/24 dev vmbr0 proto kernel scope link src 192.168.100.11
```

192.168.100.10 et 11 : l'adresse IP des machines

192.168.100.0 : l’adresse réseau

Pour la dernière partie, on devait changer le pont bridge vmbr0 et ajouter vmbr1.

Pour modifier les ponts bridge et les interfaces réseau sur l’hôte Proxmox, faut faire nano */etc/network/interfaces* et modifier les informations nécessaires pour la suite :

Bridge : Permet de créer des ponts, des réseaux virtuels et agréger des interfaces réseaux. Un pont permet de connecter le réseau virtuel de Proxmox VE au réseau local physique.

Vmbr0 pour les deux VM : 192.168.100.1

Vmbr1 pour les deux VM : 192.168.200.1

Avant	Après

`auto lo`
`iface lo inet loopback`
 	
`iface nic0 inet manual`
 	
`auto vmbr0`
`iface vmbr0 inet static`
`address 192.168.100.10/24`
`gateway 192.168.100.1`
`bridge-ports nic0`
`bridge-stp off`
`bridge-fd 0`
 	
`auto vmbr1`
`iface vmbr1 inet static`
`address 192.168.200.1`
`netmask 255.255.255.0`
`bridge_ports none`
`bridge_stp off`
`bridge_fd 0_`
 	
`iface nic1 inet manual`
 	
`source /etc/network/interfaces.d/*`
	`auto lo`
	`iface lo inet loopback`
	
	`iface nic0 inet manual`
	
	`auto vmbr0`
	`iface vmbr0 inet static`
	`address 192.168.100.1`
	`netmask 255.255.255.0`
	`bridge-ports nic0`
	`bridge-stp off`
	`bridge-fd 0`
	
	`auto vmbr1`
	`iface vmbr1 inet static`
	`address 192.168.200.1`
	`netmask 255.255.255.0`
	`bridge_ports none`
	`bridge_stp off`
	`bridge_fd 0`
	
	`iface nic1 inet manual`
	
	`source /etc/network/interfaces.d/*`

Cluster : connecte deux ou plusieurs ordinateurs via un réseau afin qu'ils fonctionnent de manière cohérente comme un seul et même système. En général, une configuration en cluster se compose de nœuds de calcul, d'un nœud principal, d'un équilibreur de charge et d'un mécanisme de pulsation.

Résultat des statuts des clusters : *pvecm status*

```
root@pve1:~# pvecm status
Cluster information
-------------------
Name:             cluster-bts
Config Version:   1
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Jan 19 10:56:47 2026
Quorum provider:  corosync_votequorum
Nodes:            1
Node ID:          0x00000001
Ring ID:          1.19
Quorate:          Yes

Votequorum information
----------------------
Expected votes:   1
Highest expected: 1
Total votes:      1
Quorum:           1
Flags:            Quorate

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.100.10 (local)
```

```
root@pve2:~# pvecm status
Cluster information
-------------------
Name:             cluster-bts
Config Version:   1
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Mon Jan 19 11:01:13 2026
Quorum provider:  corosync_votequorum
Nodes:            1
Node ID:          0x00000001
Ring ID:          1.1e
Quorate:          Yes

Votequorum information
----------------------
Expected votes:   1
Highest expected: 1
Total votes:      1
Quorum:           1
Flags:            Quorate

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.100.11 (local)
```

Résultats des statuts des environnements des clusters : *systemctl status pve-cluster*

```
root@pve1:~# systemctl status pve-cluster
* pve-cluster.service - The Proxmox VE cluster filesystem
     Loaded: loaded (/usr/lib/systemd/system/pve-cluster.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-01-20 15:05:15 CET; 43min ago
 Invocation: eb6997c337234d12aac865ae403c2a99
    Process: 832 ExecStart=/usr/bin/pmxcfs (code=exited, status=0/SUCCESS)
   Main PID: 852 (pmxcfs)
      Tasks: 7 (limit: 9366)
     Memory: 58.9M (peak: 69.3M)
        CPU: 2.180s
     CGroup: /system.slice/pve-cluster.service
             └─852 /usr/bin/pmxcfs

Jan 20 15:08:58 pve1 pmxcfs[852]: [status] notice: received sync request (epoch 1/852/00000002)
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: received all states
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: leader is 1/852
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: synced members: 1/852
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: start sending inode updates
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: sent all (38) updates
Jan 20 15:08:58 pve1 pmxcfs[852]: [dcdb] notice: all data is up to date
Jan 20 15:08:58 pve1 pmxcfs[852]: [status] notice: received all states
Jan 20 15:08:58 pve1 pmxcfs[852]: [status] notice: all data is up to date
Jan 20 15:09:02 pve1 pmxcfs[852]: [status] notice: received log
```

Résultat du ping :

```
root@pve1:~# ping 192.168.100.11
PING 192.168.100.11 (192.168.100.11) 56(84) bytes of data.
64 bytes from 192.168.100.11: icmp_seq=1 ttl=64 time=0.799 ms
64 bytes from 192.168.100.11: icmp_seq=2 ttl=64 time=0.688 ms
64 bytes from 192.168.100.11: icmp_seq=3 ttl=64 time=0.512 ms
64 bytes from 192.168.100.11: icmp_seq=4 ttl=64 time=0.684 ms
64 bytes from 192.168.100.11: icmp_seq=5 ttl=64 time=0.491 ms
64 bytes from 192.168.100.11: icmp_seq=6 ttl=64 time=0.371 ms
64 bytes from 192.168.100.11: icmp_seq=7 ttl=64 time=0.284 ms
```

```
root@pve2:~# ping 192.168.100.10
PING 192.168.100.10 (192.168.100.10) 56(84) bytes of data.
64 bytes from 192.168.100.10: icmp_seq=1 ttl=64 time=0.744 ms
64 bytes from 192.168.100.10: icmp_seq=2 ttl=64 time=0.664 ms
64 bytes from 192.168.100.10: icmp_seq=3 ttl=64 time=0.290 ms
64 bytes from 192.168.100.10: icmp_seq=4 ttl=64 time=0.536 ms
64 bytes from 192.168.100.10: icmp_seq=5 ttl=64 time=0.520 ms
64 bytes from 192.168.100.10: icmp_seq=6 ttl=64 time=0.419 ms
64 bytes from 192.168.100.10: icmp_seq=7 ttl=64 time=0.605 ms
```

Résultat du status après ajout de node (nœud) :

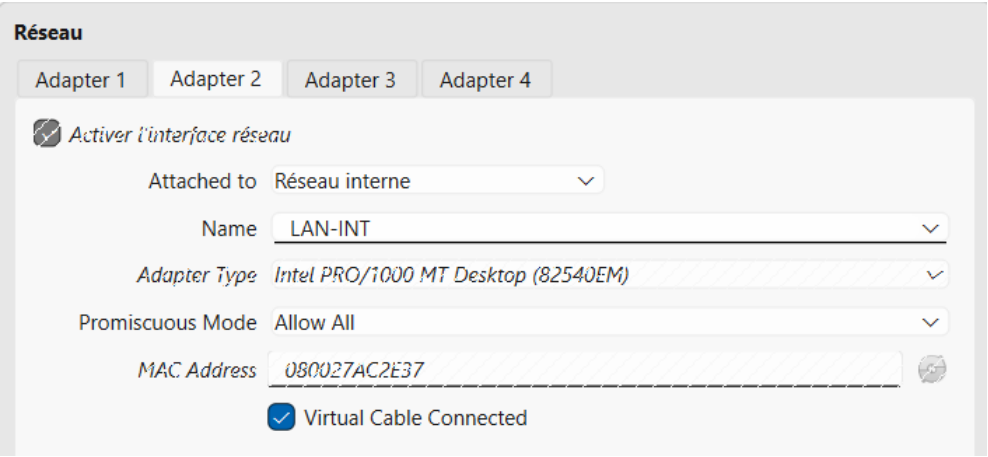
```
root@pve1:~# pvecm status
Cluster information
-------------------
Name:             cluster-bts
Config Version:   2
Transport:        knet
Secure auth:      on

Quorum information
------------------
Date:             Tue Jan 20 15:09:14 2026
Quorum provider:  corosync_votequorum
Nodes:            2
Node ID:          0x00000001
Ring ID:          1.2c
Quorate:          Yes

Votequorum information
----------------------
Expected votes:   2
Highest expected: 2
Total votes:      2
Quorum:           2
Flags:            Quorate

Membership information
----------------------
    Nodeid      Votes Name
0x00000001          1 192.168.100.10 (local)
0x00000002          1 192.168.100.11
```

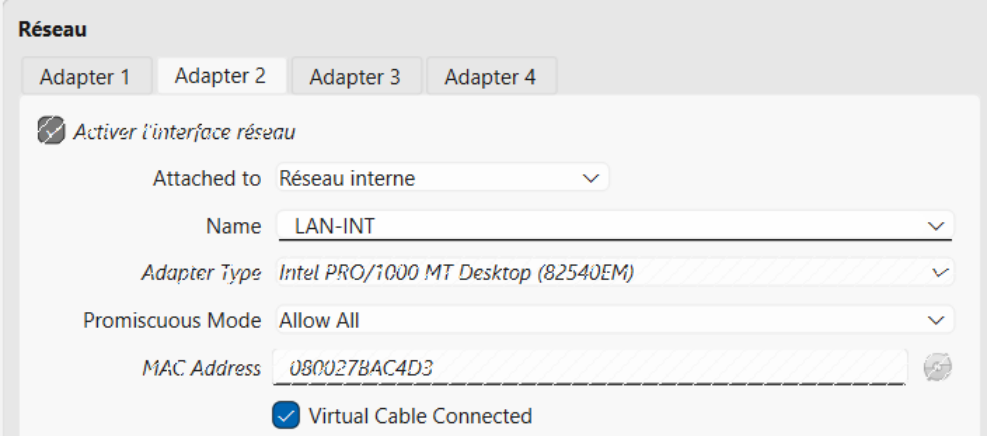
Le problème était qu'il fallait inverser les interfaces dans */etc/network/interfaces*

Avant	Après

`auto lo`
`iface lo inet loopback`
 	
`iface nic0 inet manual`
 	
`auto vmbr0`
`iface vmbr0 inet static`
`address 192.168.100.10/24`
`gateway 192.168.100.1`
`bridge-ports nic0`
`bridge-stp off`
`bridge-fd 0`
 	
`auto vmbr1`
`iface vmbr1 inet static`
`address 192.168.200.1`
`netmask 255.255.255.0`
`bridge_ports none`
`bridge_stp off`
`bridge_fd 0_`
 	
`iface nic1 inet manual`
 	
`source /etc/network/interfaces.d/*`
	`auto lo`
	`iface lo inet loopback`
	
	`iface nic0 inet manual`
	
	`auto vmbr0`
	`iface vmbr0 inet static`
	`address 192.168.100.11`
	`netmask 255.255.255.0`
	`bridge-ports nic1`
	`bridge-stp off`
	`bridge-fd 0`
	
	`auto vmbr1`
	`iface vmbr1 inet static`
	`address 192.168.200.22`
	`netmask 255.255.255.0`
	`bridge_ports none`
	`bridge-stp off`
	`bridge_fd 0`
	
	`iface nic1 inet manual`
	
	`source /etc/network/interfaces.d/*`

Puis fallait activer le mode promiscuité dans VirtualBox (mode qui se réfère à une configuration de la carte réseau, qui permet à celle-ci d'accepter toutes les trames qu'elle reçoit, même si ceux-ci ne lui sont pas adressés) pour les deux VM.





Pour les deux VM de l'interface du réseau interne

Pour le réseau, il faut modifier le bridge vmbr1 pour les deux VM sur */etc/network/interfaces*

```
auto vmbr1
iface vmbr1 inet manual
        bridge_ports nic0
        bridge-stp off
        bridge_fd 0
```

Résultat final :

```
root@pve1:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr1 state UP group default qlen 1000
    link/ether 08:00:27:a3:5a:4f brd ff:ff:ff:ff:ff:ff
    altname enx080027a35a4f
3: nic1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 08:00:27:ac:2e:37 brd ff:ff:ff:ff:ff:ff
    altname enx080027ac2e37
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:ac:2e:37 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feac:2e37/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
5: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:a3:5a:4f brd ff:ff:ff:ff:ff:ff
    inet6 fd17:625c:f037:2:a00:27ff:fea3:5a4f/64 scope global dynamic mngtmpaddr proto kernel_ra
       valid_lft 86169sec preferred_lft 14169sec
    inet6 fe80::a00:27ff:fea3:5a4f/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

```
root@pve2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr1 state UP group default qlen 1000
    link/ether 08:00:27:05:6b:14 brd ff:ff:ff:ff:ff:ff
    altname enx080027056b14
3: nic1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 08:00:27:ba:c4:d3 brd ff:ff:ff:ff:ff:ff
    altname enx080027bac4d3
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:ba:c4:d3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.11/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feba:c4d3/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
5: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 08:00:27:05:6b:14 brd ff:ff:ff:ff:ff:ff
    inet6 fd17:625c:f037:2:a00:27ff:fe05:6b14/64 scope global dynamic mngtmpaddr proto kernel_ra
       valid_lft 86253sec preferred_lft 14253sec
    inet6 fe80::a00:27ff:fe05:6b14/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```